

Exercise 4

Fix $n > 2$. Construct a random vector $(X_1, \dots, X_n)$ defined on some probability space such that (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \dots, X_n)$ is not exchangeable.

Solution Let (X, Y) be a jointly normal random vector. It is exchangeable if and only if $\mathbb{E}[X] = \mathbb{E}[Y]$ and $\text{Var}(X) = \text{Var}(Y)$. With this in mind, let $(X_1, \dots, X_n)$ be a jointly normal random vector such that $\mathbb{E}[X_i] = \mathbb{E}[X_j]$ and $\text{Var}(X_i) = \text{Var}(X_j)$ for all $i \neq j$ but $\text{Cov}(X_1, X_2) \neq \text{Cov}(X_1, X_3)$. Then (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \dots, X_n)$ is not exchangeable. For a concrete example, say $\mathbb{E}[X_i] = 0$ and $\text{Var}(X_i) = 1$ for all i , $\text{Cov}(X_1, X_2) = -1$, and all other covariances are zero. ///